MODULE 53 · DOMAIN 1 - ARCHITECTURE BUILT FOR THE CLOUD

Practice Exam: Domain 1

Fifteen original questions on architecture built for the cloud. Pause, decide, then watch.

DOMAIN 1 of 5 EXAM WEIGHT 25% SET 1 of 6

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How to use this set

- **Allow 2.4 minutes per question**: 36 minutes for the set.
- **Write your answer down** before you unpause.
- **Mark the ones you guessed**, even when the guess was right.
- The last slide maps every question to the module that teaches it.

> ON THE EXAM
>
> A correct guess is a gap you have not closed yet. Count it as a miss.

> An online retailer publishes order events. Three teams consume them independently: fraud scoring, analytics and fulfilment.
>
> After a bug, the analytics team must be able to reprocess the previous seven days of events without affecting the other teams.

Which service should carry the events?

A Amazon Kinesis Data Streams with retention extended to seven days

B An SQS FIFO queue shared by all three teams

C An SNS topic with an SQS queue per team

D EventBridge with a rule per team

◆ PAUSE AND DECIDE

Which service should carry the events?

A Kinesis, 7-day retention

Independent readers, each able to replay history.

B Shared FIFO queue

One worker pool; messages leave once consumed.

C SNS fan-out to SQS

Fan-out works, but deleted messages cannot be replayed.

D EventBridge rules

Routes events; no seven-day replay per consumer.

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Several independent readers plus replay: a stream. Module 01.

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> A lender's loan workflow calls a credit bureau, then waits for an underwriter to approve. Approval can take up to five days, and the step must escalate if nobody responds in 48 hours.
>
> The team wants the least custom code.

What should the solutions architect use?

A. A Lambda function that polls a DynamoDB table for the approval

B. Step Functions Express workflow with a Wait state

C. Step Functions Standard workflow with a task token, a timeout and a Catch that escalates

D. An SQS delay queue that re-checks every 15 minutes

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Polling Lambda
Custom polling code and state.

B Express workflow
Express runs last at most five minutes; no callbacks.

C Standard, task token
Waits days for a callback; timeout drives escalation.

D SQS delay queue
Custom re-check loop.

THE SIGNAL

Waiting days for a person: Standard with a task token. Module 02.

A media company runs EKS with 14 managed node groups, one per instance type. Pods wait several minutes for capacity during spikes, and nodes are often half empty.

The platform team wants faster scale-out and better packing, with as little node management as possible.

What should the solutions architect recommend?

A. Add more node groups with larger instance types

B. EKS Auto Mode, which manages Karpenter-based provisioning for the cluster

C. Increase the Cluster Autoscaler scan interval

D. Move all workloads to Fargate profiles

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A More node groups

More sprawl, same slow scale-out.

B EKS Auto Mode

Fast, right-sized nodes; AWS manages provisioning.

C Tune Cluster Autoscaler

Tuning does not fix group-based bin-packing.

D All Fargate

Removes nodes but loses DaemonSets, GPUs, some features.

THE SIGNAL

Slow scale-out, poor packing, node group sprawl: Karpenter, or Auto Mode to have it managed. Module 03.

A payments API runs behind an Application Load Balancer that uses path-based routing. A bank partner will only connect to two fixed IP addresses on its firewall allow-list.

The ALB's routing rules must stay as they are.

What should the solutions architect add?

A Replace the ALB with a Network Load Balancer

B Elastic IP addresses attached to the ALB

C A CloudFront distribution in front of the ALB

D AWS Global Accelerator with the ALB as its endpoint

◆ PAUSE AND DECIDE

What should the solutions architect add?

A Replace with NLB

Loses the ALB's path-based routing.

B Elastic IPs on ALB

An ALB cannot take Elastic IP addresses.

C CloudFront

CloudFront IP addresses change.

D Global Accelerator

Two static anycast IPs, ALB unchanged.

THE SIGNAL

Fixed IPs, keep the ALB: Global Accelerator in front of it. Module 04.

A SaaS provider stores every tenant's documents in one S3 bucket under tenant prefixes. A large tenant requires the ability to revoke the provider's access to its data at any time, immediately.

Other tenants have no such requirement.

What should the solutions architect design for that tenant?

A. A separate prefix with a stricter bucket policy

B. SSE-S3 encryption with versioning

C. Encrypt that tenant's objects with a KMS key owned by the tenant's account, which the tenant can disable

D. A separate IAM role for the tenant's data

◆ PAUSE AND DECIDE

What should the solutions architect design for that tenant?

A Stricter prefix policy

The provider controls the policy, so can change it.

B SSE-S3

AWS-managed key; the tenant controls nothing.

C Tenant-owned KMS key

Tenant disables the key; data becomes unreadable.

D Separate IAM role

The provider controls the role.

THE SIGNAL

A tenant who must be able to revoke access needs a key it controls. Module 05.

A European insurer uses Amazon Bedrock in eu-central-1. Peak traffic now exceeds the Region's on-demand throughput for its model.

Regulation requires that prompts and responses are processed only within the EU.

What should the solutions architect configure?

A. A geographic cross-Region inference profile for the EU

B. A global cross-Region inference profile

C. Call the model in us-east-1 when eu-central-1 throttles

D. Purchase Provisioned Throughput in us-west-2

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A EU geographic profile
Spreads load across EU Regions only.

B Global profile
May route outside the EU.

C Fail over to us-east-1
Processes data in the US.

D Provisioned in us-west-2
Processes data in the US.

THE SIGNAL

Residency plus more throughput: a geographic cross-Region inference profile. Module 06.

A publisher wants retrieval-augmented generation over 60 million archived articles. Queries are rare, a few hundred a day, and a second of latency is acceptable.

Storage cost for the vectors must be as low as possible, with a managed Bedrock knowledge base.

Which vector store should the knowledge base use?

A Amazon OpenSearch Serverless

B Aurora PostgreSQL with pgvector

C Amazon Neptune Analytics

D Amazon S3 Vectors

◆ PAUSE AND DECIDE

Which vector store should the knowledge base use?

A OpenSearch Serverless

Fast and busy-friendly; costly for cold scale.

B Aurora pgvector

A database to size and run for rare queries.

C Neptune Analytics

Graph analytics, not the cost-optimised choice.

D S3 Vectors

Lowest-cost store for large, rarely queried vectors.

THE SIGNAL

Huge and rarely queried: S3 Vectors. Busy and interactive: OpenSearch. Module 07.

A manufacturer must connect its data centre to AWS within one week for a migration pilot. Within three months it will need a steady 5 Gbps with predictable latency.

Connectivity must survive the failure of either path.

What should the solutions architect plan?

A Order Direct Connect now and wait for it before starting the pilot

B Site-to-Site VPN now; add Direct Connect later and keep the VPN as backup

C Two Site-to-Site VPN connections as the permanent design

D A single 10 Gbps Direct Connect connection with no backup

◆ PAUSE AND DECIDE

What should the solutions architect plan?

A Wait for Direct Connect
Misses the one-week deadline.

B VPN now, DX later, VPN backup
Fast start, then steady throughput with resilience.

C VPN permanently
No predictable latency; VPN tunnels cap throughput.

D Single DX
One path; fails the resilience requirement.

THE SIGNAL

Needed this week: VPN. Sustained and predictable: Direct Connect. Most answers use both. Module 08.

> Applications in a corporate data centre must resolve names in a Route 53 private hosted zone associated with a shared-services VPC. The data centre is connected by Direct Connect.
>
> On-premises DNS servers run Windows Server.

What should the solutions architect configure?

A. A Route 53 VPC Resolver inbound endpoint, with on-premises conditional forwarders pointing to it

B. A Route 53 VPC Resolver outbound endpoint with a forwarding rule

C. Make the hosted zone public

D. Copy the records into on-premises DNS each night

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Inbound endpoint

On premises asks; inbound receives the queries.

B Outbound endpoint

For AWS resolving on-premises names.

C Public zone

Exposes internal names.

D Nightly copy

Stale records, manual work.

THE SIGNAL

Who needs the answer decides the endpoint: on premises asking means inbound. Module 09.

A logistics group operates in 12 AWS Regions. Every Region needs the same three network segments, production, development and shared services, with production isolated from development.

The network team wants one policy document to define the global network, not separate route tables per Region.

What should the solutions architect use?

A. A Transit Gateway in each Region with peering and per-Region route tables

B. VPC peering between every pair of VPCs

C. AWS Cloud WAN with a core network policy defining the segments

D. A single Transit Gateway in one Region

◆ PAUSE AND DECIDE

What should the solutions architect use?

A TGW per Region, peered

Works, but segments managed per Region by hand.

B Full-mesh peering

Unmanageable at this scale.

C Cloud WAN

One policy defines segments in every Region.

D One Transit Gateway

Transit Gateway is regional.

THE SIGNAL

Many Regions, same segments, one policy: Cloud WAN. Module 10.

A bank must inspect all traffic between VPCs and to the internet using a specific third-party firewall product. It already owns licences for that product.

The design must scale the appliances horizontally and keep flows symmetric.

What should the solutions architect use?

A AWS Network Firewall in a central inspection VPC

B Security groups and network ACLs only

C AWS WAF on every load balancer

D A Gateway Load Balancer in front of the third-party appliances in an inspection VPC

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Network Firewall

Managed, but not the named third-party product.

B SGs and NACLs

Not deep inspection.

C AWS WAF

Web requests only, not all traffic.

D Gateway Load Balancer

Scales third-party appliances, keeps flows symmetric.

THE SIGNAL

A named third-party firewall or existing licences: Gateway Load Balancer. Module 11.

A trading platform writes account balances in two Regions. If a Region fails, no committed write may be lost, and a read in either Region must return the latest balance.

The team already uses DynamoDB.

What should the solutions architect configure?

A DynamoDB global tables with eventual consistency

B DynamoDB global tables with multi-Region strong consistency

C A single-Region table with cross-Region backups

D DynamoDB Streams replicating to a second table by Lambda

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Global tables, eventual

Recent writes can be lost; reads can be stale.

B Global tables, MRSC

RPO zero, latest read anywhere.

C Backups

RPO of the backup interval.

D Streams and Lambda

Asynchronous, custom, lossy.

THE SIGNAL

RPO zero and read-your-latest-write in any Region: MRSC. Chosen at creation. Module 12.

A retailer's data lake holds customer orders as Parquet files in S3. Under GDPR, it must delete individual customers' rows within days of a request, and nightly jobs upsert changed orders.

The data team does not want to run compaction or snapshot clean-up jobs.

What should the solutions architect recommend?

A Rewrite whole Parquet partitions for every deletion

B Move the lake into Amazon Redshift

C Store the tables as Apache Iceberg in Amazon S3 Tables

D Self-managed Apache Iceberg tables with scheduled maintenance jobs

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Rewrite partitions

Slow, error-prone, conflicts with writers.

B Move to Redshift

A large change for a table-format problem.

C Iceberg in S3 Tables

Row-level deletes and upserts, maintenance managed.

D Self-managed Iceberg

Works, but runs the maintenance they want to avoid.

THE SIGNAL

Row-level deletes or upserts on the lake: Iceberg. Without running maintenance: S3 Tables. Module 13.

A healthcare company's services run on EC2, ECS and Lambda across 25 VPCs in eight accounts. Some VPCs have overlapping CIDR ranges.

Each service-to-service call must be authorised with IAM, and teams want to avoid managing routing between VPCs.

What should the solutions architect recommend?

A Amazon VPC Lattice service networks shared with AWS RAM, with IAM auth policies

B Transit Gateway with re-addressed VPCs

C ECS Service Connect across all services

D A PrivateLink endpoint service for every service

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A VPC Lattice
Mixed compute, overlapping CIDRs, IAM per request.

B Transit Gateway
Re-addressing work, no per-request IAM.

C Service Connect
ECS only.

D PrivateLink per service
Hundreds of endpoints, no shared auth model.

THE SIGNAL

Mixed compute, crossing VPCs, or per-request IAM authorisation: VPC Lattice. Module 04.

> A subscription service processes renewals with Lambda functions reading from an SQS FIFO queue with content-based deduplication. Occasionally a function charges a card and then times out, and the message is processed again.
>
> Some customers have been charged twice.

What should the solutions architect change?

A Increase the deduplication interval

B Make the charge idempotent by recording a renewal ID with a conditional write before charging, and checking it on retry

C Switch to a standard SQS queue

D Set the maximum receive count to one

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Longer dedup window

The deduplication interval is fixed, and it stops duplicate sends only.

B Idempotent consumer

A retry finds the record and skips the charge.

C Standard queue

Adds duplicates rather than removing them.

D Max receives one

Sends failures to the dead-letter queue, losing renewals.

THE SIGNAL

Broker deduplication stops duplicate sends. Only an idempotent consumer stops duplicate effects. Module 01.

Where to revisit

IF YOU MISSED	REWATCH
Q1, Q15	Module 01: Event-driven architecture
Q2	Module 02: Stateless design and durable workflows
Q3	Module 03: Containers and orchestration
Q4, Q14	Module 04: Service mesh and ingress
Q5	Module 05: Edge and multi-tenant isolation
Q6, Q7	Modules 06-07: Bedrock, RAG and inference
Q8, Q9	Modules 08-09: Hybrid connectivity and DNS
Q10, Q11	Modules 10-11: Multi-VPC and network security
Q12, Q13	Modules 12-13: Databases and data lakes

ON THE EXAM

12 or more out of 15 is on track for this domain. Below 9, rewatch before the next set.